

CariSurg MedTech Pathways Research Project

Preliminary Proposal

Project Title:

Explainable AI-Assisted Emergency Department Triage Support System for Resource-Constrained Healthcare Settings

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Statement of the Problem

Emergency Department (ED) triage aims to prioritize patients according to urgency, but decisions occur across multiple workflow stages where information becomes available gradually rather than simultaneously. Registration, vital sign collection, nurse assessment, and queue management each introduce opportunities for delay, incomplete information, and communication loss. As a result, prioritization decisions may occur before the full clinical picture is available. These workflow realities can reduce consistency and increase operational pressure. Therefore, any AI-assisted triage solution must improve decision support while fitting naturally into existing clinical processes.

Proposed Solution

This project proposes an AI-assisted clinical decision support system integrated into the Emergency Department triage workflow. Rather than replacing nurse judgement, the system would analyze information as it becomes available throughout registration, vital sign collection, and triage assessment to provide contextual support. The proposed system would identify abnormal observations, surface risk indicators, and support prioritization decisions while preserving clinician autonomy. It would operate passively alongside existing processes to minimize additional documentation and workflow disruption. By continuously updating recommendations as new information becomes available, the system aims to support timely assessment while respecting operational realities and maintaining human oversight.

Literary Review

[1] This review examined how artificial intelligence is being applied to emergency department triage to reduce overcrowding and improve patient prioritization. AI was primarily used through machine learning and natural language processing approaches that processed patient data to support triage decisions and optimize resource allocation. The review reported improvements in wait times, reductions in time-to-treatment, and more efficient bed and staffing allocation. However, the authors identified challenges including algorithmic bias, clinician distrust, and concerns surrounding black-box decision-making.

[2] This scoping review explored the use of artificial intelligence across emergency department triage systems to improve patient classification and outcome prediction. AI was applied using machine learning algorithms trained on patient demographics, symptoms, and clinical indicators to support prioritization decisions. The review found that AI systems frequently outperformed traditional triage approaches, with some models demonstrating substantially improved predictive performance. The authors highlighted limitations related to retrospective datasets, inconsistent preprocessing methods, and limited external validation.

[3] This study developed TriageIntelli, an AI-assisted triage system designed to support emergency assessment and reduce delays caused by increasing patient volumes. Artificial intelligence was implemented through six supervised machine learning models trained on emergency department data and combined into a stacked ensemble architecture. The final stacked model achieved the highest performance, reaching 80.05% classification accuracy and an AUROC of 0.93. Despite strong performance, the study was limited by its reliance on historical data collected from only two Korean emergency departments.

[4] This study investigated whether machine learning and natural language processing could improve emergency department triage by predicting patient clinical disposition. AI was used to analyze structured triage data together with free-text clinical notes to identify patterns associated with admission outcomes. The results demonstrated improved predictive capability compared with conventional approaches and highlighted the value of combining structured and unstructured clinical information. Limitations included challenges associated with model generalization and dependence on local documentation practices.

[5] This comparative study evaluated the performance of large language models in emergency medicine triage tasks relative to inexperienced human decision-makers. AI was implemented using large language models to generate triage recommendations across simulated emergency scenarios. The findings demonstrated promising performance and highlighted the potential of language models to support triage workflows and decision making. However, concerns remained regarding reliability, transparency, and the safe deployment of generative AI systems in clinical environments.

[6] This study developed and validated a machine learning model to predict 72-hour emergency department return visit admissions (RVA) using electronic health record data from multiple urban hospitals. Several algorithms, including logistic regression, gradient boosting methods, and a clustering-based approach, were evaluated to identify high-risk patients. The final model achieved strong predictive performance in the development dataset (AUC up to 0.87) but showed reduced performance in external validation (AUC 0.75), highlighting challenges in generalizability across clinical settings. The study also incorporated interpretability techniques such as SHAP analysis and clinician chart review to assess model

reasoning and clinical relevance. While the model demonstrated potential for identifying patients at risk of deterioration after discharge, the authors emphasized that limited clinical actionability, dataset imbalance, and variation in hospital workflows restrict real-world implementation without further refinement and external validation.

[7] This study developed an interpretable machine learning emergency department triage model designed to improve prediction performance while addressing class imbalance in emergency datasets. Artificial intelligence was implemented through an interpretable ML framework that generated transparent triage recommendations and prioritization scores rather than relying on black-box outputs. The study reported improved triage performance and demonstrated that explainable approaches may strengthen clinician trust and adoption. Limitations included dependence on local population characteristics and uncertainty regarding transferability across different emergency healthcare settings.

[8] This study addressed inefficiencies in patient flow within a Caribbean emergency department. The authors used qualitative workflow analysis to examine registration, communication, and clinical coordination processes. The findings identified bottlenecks caused by information delays, workload pressures, and communication breakdowns. However, the study was limited by its qualitative design and lack of quantitative performance metrics, limiting generalizability.

[9] This study addressed the problem of improving triage accuracy and outcome prediction compared to traditional methods. The authors used machine learning models trained on electronic triage data to predict clinical outcomes and prioritization levels. The results showed that machine learning improved differentiation of high-risk patients compared to conventional triage systems. However, limitations included challenges in integration into clinical workflows and the need for clinician oversight in decision-making.

[10] This study addressed the challenge of implementing clinical decision support systems effectively in healthcare environments. The authors used an evaluative approach examining adoption factors such as usability, workflow integration, and clinician training. The findings showed that successful implementation depends heavily on system design, trust, and alignment with clinical workflows. However, limitations included variability across healthcare settings and the difficulty of standardizing adoption conditions.

[11] This systematic review examined why transparency is considered an ethical requirement for artificial intelligence based clinical decision support systems in healthcare. The authors reviewed published literature discussing accountability, trust, fairness, patient autonomy, and explainability to identify the ethical principles supporting AI adoption in clinical practice. The review found that transparent AI systems strengthen clinician trust, improve accountability, and promote safer integration of AI into healthcare. However, the authors noted that many ethical arguments remain theoretical and highlighted the lack of standardized methods for evaluating how transparent AI systems should be.

[12] This study explored the role of transparent artificial intelligence in medical decision-making and its influence on clinician trust and patient safety. The authors examined existing approaches that help healthcare professionals understand how AI-generated recommendations are produced and interpreted during clinical practice. The findings suggested that interpretable systems improve clinician confidence, encourage informed decision making, and promote responsible implementation while maintaining human oversight. However, the

authors acknowledged that evidence from real world healthcare settings remains limited and that further validation is needed before widespread adoption.

[13] This article investigated whether clinicians should understand how artificial intelligence reaches its recommendations in order to support safe and ethical healthcare decision making. The authors critically examined current evidence alongside ethical and regulatory perspectives surrounding AI in clinical practice. The study concluded that greater transparency improves clinician understanding, supports appropriate oversight, and strengthens confidence in AI-assisted decision making, but emphasized that transparency alone cannot guarantee safe or accurate clinical outcomes. The authors also noted that limited empirical evidence currently exists demonstrating the direct impact of these approaches on patient outcomes.

[14] This review examined the application of interpretable artificial intelligence in healthcare and the challenges associated with implementing transparent machine learning models in clinical settings. The authors surveyed a range of techniques designed to make AI predictions easier for clinicians to understand and compared their use across different healthcare applications. The review found that these methods improve model transparency, support clinician understanding, and encourage greater trust in AI assisted decision support systems. However, the authors highlighted that no universally accepted framework currently exists for evaluating these techniques and that many remain difficult to apply consistently across different clinical environments.

[15] This study investigated how transparent artificial intelligence can improve trust in AI assisted healthcare decisions while addressing concerns related to accountability and patient safety. The authors reviewed existing research on methods that help clinicians understand the reasoning behind AI recommendations and analyzed how these approaches influence confidence in AI supported decision making. The findings suggested that greater transparency promotes responsible AI adoption and supports safer clinical decisions by making recommendations easier to interpret. Nevertheless, the authors acknowledged that balancing predictive performance with model interpretability remains challenging and emphasized the need for additional real-world clinical evaluations.

Healthcare Workflow

Patient Arrival



Registration

(Administrative details collected; patient reason for visit recorded)



Vitals Collection

(Vital signs measured and temporarily held until triage assessment)



Triage Nurse Assessment

(Chief complaint + clinical judgement + urgency assignment)



Priority Assignment

(Patient enters queue based on assigned urgency)



Waiting Area Monitoring

(Patient condition may change while waiting)



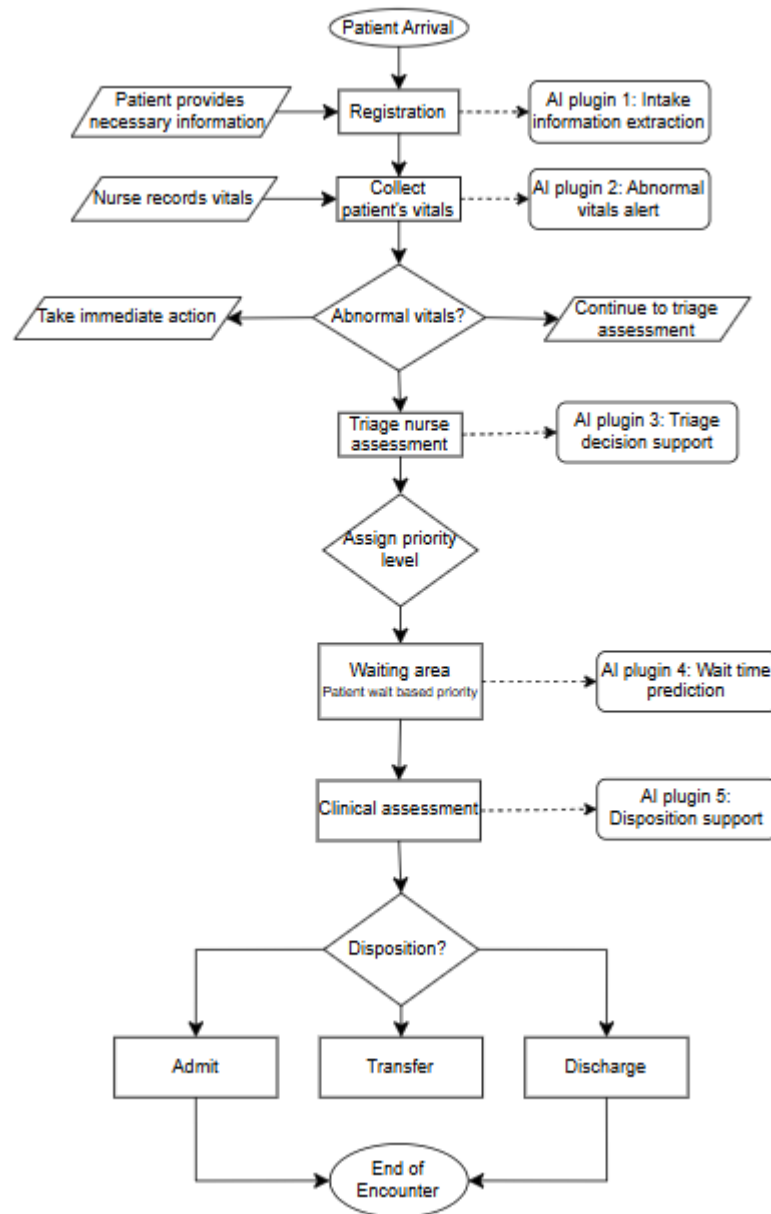
Clinician Assessment



Disposition

(Discharge / Admit / Transfer)

Information enters progressively throughout the workflow and clinical decisions are revised as additional information becomes available.



AI Integration Points

AI Plug-in 1 - Registration Support

Purpose: Structure intake information into standard fields before assessment begins.

AI Plug-in 2 - Observation Validation

Purpose: Detect missing or unusual combinations of vital signs before triage completion.

AI Plug-in 3 - Triage Decision Support

Purpose: Present risk indicators alongside nurse assessment rather than replacing categorization.

AI Plug-in 4 - Queue Monitoring

Purpose: Identify patients whose wait duration and observations suggest reassessment.

AI Plug-in 5 - Disposition Awareness

Purpose: Support clinicians by surfacing relevant outcome indicators.

Constraints

Constraint 1: Sequential Information Capture Creates an Early Decision Blind Spot

Workflow observation:

Registration occurs before full observations and assessment are available.

Mechanism:

Initial prioritization decisions may begin before complete vital signs and clinical context are captured.

AI implication:

The system must update recommendations dynamically rather than assuming complete data availability.

Design response:

Display confidence levels and continuously refresh outputs.

Measurement:

Percentage of recommendation changes after vital completion.

Constraint 2: Multiple Handoffs Create Information Translation Risk

Workflow observation:

Patient information moves across registration staff, triage nurses, waiting stages, and clinicians.

Mechanism:

Information gathered early may not remain visible or equally emphasized later.

AI implication:

The system must preserve context across transitions.

Design response:

Carry forward structured summaries instead of requesting repeated entry.

Measurement:

Number of repeated data collection events per patient.

Constraint 3: Queue Status Does Not Automatically Trigger Reassessment

Workflow observation:

Patients wait according to assigned urgency while clinical condition may evolve.

Mechanism:

A patient may deteriorate after triage without triggering reassessment.

AI implication:

AI should monitor elapsed waiting time together with available observations.

Design response:

Generate reassessment prompts rather than reassign urgency automatically.

Measurement:

Time between deterioration indicators and reassessment.

Stakeholders

Triage Nurse

Concern: Maintaining assessment speed without duplicate documentation.

Emergency Physician

Concern: Receiving reliable prioritization support.

Patient

Concern: Fair and timely assessment.

Hospital Administration

Concern: Improving patient flow without increasing workload.

Clinical IT Team

Concern: Integration into existing infrastructure.

Reflection

The initial proposal focused primarily on improving triage prediction. Workflow mapping shifted the focus toward understanding how information moves through the Emergency Department and where delays emerge. This highlighted that healthcare AI creates value only when it adapts to operational realities, preserves clinician judgement, and fits existing patterns of work. Model performance alone is insufficient if recommendations arrive at the wrong stage of the workflow or increase documentation burden.

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